Web Systems and Technologies 2
2nd Semester, School Year 2025–2026

PROJECT PROPOSAL

SAGANA: A Web-Based Integrated Agricultural Supply Chain, Inventory, and Digital Marketplace System for the Cooperative of Payanas, Torrijos, Marinduque

Keywords:
Agricultural Supply Chain, Digital Marketplace, MERN Stack, REST API, RBAC, Inventory Management, Cooperative, React.js

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CHAPTER 1: TECHNICAL DESCRIPTION & PROPOSAL

1.1 Background of the Study

The agricultural sector serves as the main economic support for numerous rural communities in the Philippines which includes Torrijos on Marinduque Island where cooperative farming and seasonal harvests and local trade function as the community's economic base. Although the agricultural sector holds great importance, farmers in the Payanas Cooperative of Torrijos still use manual methods and paper documents for managing their supply chains and tracking their inventory and conducting their market transactions. The current system of operations creates multiple problems including post-harvest losses and inventory errors which lead to slow decision processes and restrict market access for both farmers and cooperative members.

The development of web technologies together with cloud-based platforms creates a major opportunity to enhance agricultural work processes in cooperatives. The implementation of RESTful APIs together with real-time data dashboards establishes a communication link between producers and cooperative administrators and end consumers. The commercial solutions that currently exist in the market fail to meet the needs of Philippine agricultural cooperatives because they are either overpriced or they lack local features or they provide standard functions which do not suit specific operational needs.

The SAGANA provides a solution to this problem through its development of a MERN-stack web application which serves as the main software solution for the Payanas Cooperative of Torrijos, Marinduque. The platform provides farmers and cooperative administrators with real-time information and secure access through role-based permissions because it combines supply chain tracking and inventory management with a digital marketplace solution.

1.2 Objectives (General and Specific)

General Objective: To design, develop, and deploy a full-stack web-based Integrated Agricultural Supply Chain, Inventory, and Digital Marketplace System using the MERN stack for the Payanas Cooperative of Torrijos, Marinduque, enabling real-time inventory monitoring, streamlined supply chain operations, and direct online selling between cooperative members and consumers by the end of the current semester.

Specific Objectives:

1. To implement a secure User Authentication and Role-Based Access Control (RBAC) system which uses JWT and bcryptjs to create three access levels for Cooperative Admin and Farmer Member and Public Marketplace Buyer roles.
2. To develop a RESTful Node.js/Express API backend which enables full CRUD operations for four MongoDB collections that include Users and Products and Inventory Batches and Orders and uses Mongoose schema validation to safeguard against data corruption.
3. To build a React.js frontend with Vite that fetches real-time inventory and product data from the backend to create dashboards for different user roles and a public marketplace interface that responds to user needs.

4. To deploy the complete system (MongoDB Atlas, Node/Express backend, and React frontend) to cloud hosting platforms with zero critical build errors and a documented workflow by the semester deadline.

1.3 Review of Related Literature and Systems

The development of the SAGANA system is informed by a comprehensive evaluation of existing digital platforms and research articles published within the last five years. This review identifies the technical functionalities, operational models, and inherent limitations of current systems, providing the empirical foundation for the SAGANA architectural design.

Related System 1: eMarket (Department of Agriculture, Philippines)

The Department of Agriculture (DA) built the eMarket platform as a centralized online system which enables Filipino farmers to sell their products directly to both institutional and individual customers. The system aims to create market connections which will decrease the power of speculative traders who currently control regional trading centers. Producers can display their items through eMarket which provides them essential online presence but its digital capabilities fall short of being complete. The system lacks an integrated inventory management system which prevents farmers from tracking their current stock status and documenting their batch inventory information. The platform lacks specific administrative dashboards which would help cooperatives manage their operations because organized groups need these functions to handle shared sales and track inventory between their members. SAGANA will eliminate these deficiencies through the implementation of complete CRUD functions which enable cooperative managers to verify physical inventory and assess product listings [1]

Related System 2: FarmLink (AgriPhilippines)

FarmLink serves as a mobile-based platform that enables farmers to display their produce while receiving order requests from customers. The platform aims to create an intuitive system which enables producers with basic technical skills to use it without difficulty. The system review process shows that its supply chain capabilities are not yet complete because the architectural assessment needs to be completed. Modern consumers and institutional buyers who require food safety assurances expect companies to provide complete traceability because they want to track products through all stages of distribution. FarmLink does not have the required multi-role administrative system which constitutes a vital requirement for managing cooperative operations. The SAGANA system will enhance the existing model through its implementation of secure Role-Based Access Control (RBAC) system which uses JSON Web Tokens (JWT) and bcryptjs to establish different access rights for Cooperative Admins, Farmer Members, and Public Buyers [2][3].

Related System 3: iAmFarmer (Southeast Asia)

The iAmFarmer platform serves as a regional agricultural instrument which enables users to manage product inventories and track their orders throughout multiple Southeast Asian countries. The platform has achieved recognition due to its user interface design which matches

the shopping habits of Filipino customers who prefer visual content. The backend APIs of iAmFarmer fail to deliver sufficient support because they lack essential infrastructure needed to process modern cooperative operational requirements. The supply chain model treats Middle Mile operations as a permanent marketplace which fails to recognize all supply chain components needed for monitoring storage conditions and coordinating regional distribution activities. The design principles of iAmFarmer accessibility will guide SAGANA development which will use a RESTful Node.js/Express API to manage real-time inventory updates through its React.js frontend [4].

Related System 4: Agro-DigitalPH

Agro-DigitalPH provides a comprehensive value chain solution for farmer cooperatives through its use of social organization resources to create both operational standards and organizational objectives. The system solves market fragmentation through its "virtual aggregation" system which combines demand with production and fulfillment operations to create cost efficiencies. The platform uses "EasyasCash" as its digital currency which enables users to make payroll payments and transfer funds while maintaining transparent records of their monetary transactions. Agro-DigitalPH contains advanced functions but its main obstacles to user acceptance result from existing institutional problems instead of technical issues because small cooperatives cannot meet the platform's digital requirements and financial standards. The process of expanding such tools becomes difficult because users must deal with multiple regulatory systems throughout the island country. SAGANA provides a solution through its user-friendly interface which helps rural users by decreasing their mental workload while it handles all the specific tasks needed by Payanas Cooperatives without needing the complex system management found in large commercial systems [5].

Related System 5: TradeNet (Singapore, AGD)

TradeNet is a globally recognized trade facilitation platform utilized by the Singaporean government to automate cargo declarations and permits. Although it is not inherently an agricultural system, its multi-stakeholder access model and workflow automation provide a critical architectural reference for SAGANA. TradeNet's success is built on the synchronization of complex data streams among various actors, ensuring that regulatory oversight does not delay the flow of trade. SAGANA will utilize these concepts to design its cooperative governance module, ensuring that the approval of listings and the fulfillment of orders follow a streamlined, automated workflow that minimizes administrative overhead for the cooperative [6].

Related System 6: Mayani (Philippines)

Mayani is a full-stack agritech platform in the Philippines that connects over 144,000 smallholder farmers directly to institutional buyers, such as hotels and restaurant chains. The platform has successfully demonstrated that removing the seven layers of middlemen can increase farmer incomes by at least 30%. The primary limitation of the Mayani model is its high dependence on large-scale institutional demand and substantial catalytic capital, which may not be readily available to independent cooperatives in isolated provinces like Marinduque. Additionally, while Mayani engages cooperatives, it functions primarily as a centralized buyer rather than a tool for

the cooperative's internal governance. SAGANA addresses this by providing a MERN-stack application that is owned and managed by the local cooperative, allowing them to operate their own independent digital marketplace [7].

Related System 7: SariSuki (Philippines)

SariSuki is a social commerce platform that leverages community group buying to modernize the agricultural supply chain. Founded by former Grab Philippines executives, it enables Ka-Sari community leaders to act as local resellers of fresh produce, achieving 36x growth within its first nine months. The platform also operates Supah, a quick-commerce arm that utilizes a network of dark stores to deliver groceries in under 15 minutes in urban areas. While SariSuki excels at demand aggregation, its rapid-delivery model is highly capital-intensive and relies on dark store infrastructure that is not currently viable in the rural context of Torrijos. Furthermore, its focus is primarily on the B2C Last Mile, often overlooking the Middle Mile logistics of the source cooperative. SAGANA will focus on the internal value chain of the cooperative, ensuring that inventory is tracked from the moment of harvest, providing the foundational data required for local market fulfillment rather than high-speed urban delivery [8].

Related System 8: Anihan Technologies / AniTech (Philippines)

Anihan Technologies provides an end-to-end IoT sensor platform combined with the CropNote web application to reduce food loss through real-time monitoring. The system addresses the visibility gap where agribusinesses rely on manual records and fixed schedules instead of actual environmental conditions. The main limitation cited by Anihan is the hardware scaling constraint, as physical sensors represent a significant upfront cost for rural producers. SAGANA will adopt a software-first approach that allows cooperatives to log inventory and supply chain data manually or through simple mobile interfaces, providing a scalable digital framework that can be later integrated with IoT sensors as funding becomes available [9].

Related System 9: DeliverE (DTI / USAID, Philippines)

DeliverE is a digital agricultural logistics platform launched by the DTI and USAID to help farmers transport produce during the COVID-19 pandemic. It connects farmers, cooperatives, and MSMEs to transporters, enabling them to ship goods at a lower cost. The platform uses a blockchain-based system to integrate e-commerce and logistics applications from consolidation points to retail. A significant drawback of DeliverE is its regional concentration and its primary orientation toward B2B wholesale transactions. While it provides tools for cooperatives to visualize performance, it does not offer the granular, role-based batch management needed for individual farmer members to track their specific produce within the system. SAGANA addresses this by implementing role-specific dashboards that allow farmer members to manage their own inventory batches while the cooperative admin handles the broader logistics and order fulfillment [10].

Related System 10: e-Kadiwa (Department of Agriculture, Philippines)

The e-Kadiwa platform demonstrates the modernized version of the KADIWA program which the state introduced during the 1980s as a solution to achieve food security and maintain stable prices. The system establishes direct connections between farmers and consumers which decrease intermediary costs by approximately 20%.The program holds strategic value for the organization yet research shows that most consumers remain unaware of the digital platform while people living in remote places cannot access Kadiwa outlets.Consumers experience travel expenses that exceed the discount benefits they receive at physical stalls and online shops. SAGANA solves this through a hyper-localized cooperative model, where the digital marketplace is managed by the local Payanas Cooperative itself, ensuring that the product mix reflects regional demand and that fulfillment is handled through existing local networks [11].

Related System 11: Project SARAi (UPLB / Department of Agriculture, Philippines)

Project SARAi (Smarter Approaches to Reinvigorate Agriculture as an Industry) functions as an action-research program which employs precision technologies to mitigate climate hazards across the country. The system uses remote sensing and satellite data and crop modeling to create location-based recommendations which support nine main crop varieties. The primary limitation of SARAi is its development as a research-based decision support system which lacks operational marketplace functions. There exists a documented gap between the available technological supply and the actual capacity of grassroots organizations to implement those technologies. SAGANA will function as the commercial execution component which takes production information from SARAi and other systems to create an inventory system which handles the actual product distribution process. [12]

Related System 12: Cropital (Philippines)

Cropital operates an investment marketplace which focuses on agriculture to enable smallholder farmers to obtain low-interest working capital through crowdfunding methods. The platform solves a financing problem which prevents 70% of farmers from accessing necessary funds because they must pay high initial expenses. The platform received negative feedback from users because it offers investors a high-risk investment approach that results in minimal returns according to users who experienced complete investment loss during the pandemic and received no clear updates about farm operations. The system needs reliable real-time data tracking tools which can monitor agricultural development throughout the entire crop cycle. SAGANA incorporates this tracking by default through its supply chain tracker which records every movement of goods from harvest to fulfillment providing the digital proof required for investor confidence [13].

Related System 13: Kita (Philippines)

Kita operates by digitizing traditional trading posts and enabling direct farmer-to-buyer trade, with SM Supermarkets serving as a major investor. Smallholders face systematic exploitation because they receive only a small portion of the retail price. The main obstacle for Kita originates from the fact that most farmers do not possess the necessary facilities to dry and

store their harvested crops. Kita maintains credit scoring through its analysis of trading post information but lacks a system to help cooperatives control their communal storage through a regional inventory management system. SAGANA will implement an inventory management module that supports batch-level tracking, enabling cooperatives to optimize their storage utilization and timing of marketplace listings [14].

Related System 14: AGREA Move Food Initiative

AGREA Agricultural Systems International established The Move Food Initiative during the COVID-19 pandemic to connect urban consumers with farmers who faced harvest transportation challenges. The organization established volunteer networks to deliver and sell products through social media platforms such as Facebook and basic Google Forms which helped them solve their logistics crisis. The system processed more than 183000 kilograms of food while assisting thousands of families but its main drawback stemmed from its dependence on manual operations and outside social media platforms that did not provide complete inventory management and automatic transaction documentation. Farmers experienced difficulties when they needed to determine their product prices and collect payments because they lacked access to traditional banking services. SAGANA provides a web application which works as a complete system to help businesses track their inventory while using automated tools to establish pricing and complete order processing through its secure marketplace system which requires no manual work through social media [15].

Related System 15: LANDBANK E-Saka / Agri Digital Access

Through mobile banking and digital credit portals the LANDBANK Agri Digital Access initiative with its E-Saka components enables farmers and cooperatives to access agri-financial services which lead to better financial inclusion. The solution provides capital to farmers through its specialized financing solutions which include "Sikat Saka" and "AGRISENSO Plus" that grant producers access to low-interest loans to use for their production and mechanization needs. The system operates its primary functions through its banking and financial services but it lacks the ability to manage all the operational activities which occur throughout agricultural supply chains. The system manages the "financial mile" process but lacks essential features needed for inventory management and batch tracking and digital marketplace links which enable businesses to monitor their product movement from harvest through to final sale. SAGANA provides operational data infrastructure which connects to Landbank financial systems to track inventory batches in real time. The system generates the "digital proof" needed for creditworthiness and insurance while it controls the physical process of selling the produce [16].

1.4 Methodology & Solution

The SAGANA system development process follows the Agile methodology which implements multiple development sprints to create a final product that meets the demands of cooperative administrators and farmers. The MERN stack serves as the architectural foundation because it provides both scalability and operational benefits through its single JavaScript environment.

Database Design: MongoDB Atlas

Traditional relational databases often struggle with the variable and unstructured data common in agriculture. MongoDB's document-oriented NoSQL structure allows for a flexible schema that accommodates the following core collections:

- **Users Collection**: Stores hashed credentials and role assignments (Admin, Farmer, Buyer).
- **Products Collection**: Contains detailed metadata for crops, including origin, variety, and quality grade.
- **Inventory Batches Collection**: Tracks specific lots from harvest to sale, logging expiry dates and current stock levels.
- **Orders Collection**: Manages the transactional history, linked to both product and buyer records via MongoDB references.

The use of MongoDB Atlas ensures that data is stored in the cloud, providing the cooperative with high availability and automated backups without the need for on-site server maintenance.

Backend Logic: Node.js and Express.js

The backend follows a RESTful API pattern, which provides a standard interface for communication between the frontend and the database. Key backend modules include:

- **Authentication Module**: Utilizes JSON Web Tokens (JWT) for session management and bcryptjs for password hashing. Middleware functions enforce RBAC on all protected routes, ensuring that a Farmer Member cannot access the financial reports of the Cooperative Admin.
- **Inventory Logic**: Processes complex CRUD operations and implements strict Mongoose validation to prevent data corruption. The system will prevent an order from being created if the requested quantity exceeds the sum of active inventory batches.
- **Supply Chain Tracker**: Records every transition of a batch (e.g., Harvested to Stored to Listed), providing the end-to-end traceability required for professionalized trade.

Frontend Interface: React.js and Vite

The frontend utilizes a component-driven architecture to build a highly responsive and interactive user experience. By using Vite as the build tool, the development team ensures fast hot-module replacement and optimized builds for production.

- **Dynamic Data Fetching**: The application uses React hooks (useEffect, useState) to fetch real-time data from the backend, ensuring that marketplace availability is always synchronized with the physical inventory.
- **Role-Specific Dashboards**: Administrators gain access to analytical tools and user management, while farmer members see a simplified interface for logging their specific produce.

- **Responsive Public Marketplace**: Designed for both desktop and mobile users, allowing local consumers to browse and place orders with the same ease as a standard e-commerce site.

1.5 Target Users / Beneficiaries

Cooperative Admin

The central administrator of the cooperative organization. The user account manager performs product listing approval and system-wide inventory monitoring while generating reports and managing order fulfillment operations. The user has complete CRUD access rights to all database collections.

Farmer Member

The Payanas Cooperative recognizes individual farmers as their registered members. The system allows them to enter their harvest data and control their product inventory and create marketplace listings while monitoring their order progress. The system permits users to access only their personal information.

Public Marketplace Buyer

Consumers, local vendors, and external buyers access the public-facing marketplace. Registered Buyers can place orders track order history and communicate with the cooperative through the platform. The cooperative encourages its youth members to participate in digital activities which will help them learn technology skills that assist older members.

Validity of the project has been certified in collaboration with the Samahan ng Pagkakaisa sa Pag-unlad ng Payanas (SP3) Agriculture Cooperative of Torrijos, Marinduque. The signed certification letter is attached as Appendix D.

1.6 Significance of the Study

The integration of agricultural supply chain tracking, inventory management, and a digital marketplace represents the most viable pathway for the revitalization of the cooperatives in Torrijos, Marinduque. The evidence indicates that digital transformation is not merely a convenience but a strategic necessity in an era of high supply chain volatility and escalating climate risks. By digitizing the Middle Mile, cooperatives can drastically reduce post-harvest losses, capture a larger share of the retail price, and build the data infrastructure required for long-term financial empowerment. SAGANA provides the following direct benefits:

- **For Farmer Members**: Reduced post-harvest losses through timely inventory alerts, increased income through direct marketplace access without middlemen, and transparent tracking of their produce from farm to sale.

- **For Cooperative Admins**: Centralized, real-time visibility into all member inventories and transactions, streamlined administrative workflows, and data-driven reporting for cooperative governance.
- **For Local Consumers and Buyers**: Access to fresh, locally sourced agricultural products at fair prices through a trustworthy digital platform.
- **For the Academic Institution**: SAGANA demonstrates the practical application of MERN stack development, RESTful API design, RBAC security, and cloud deployment, synthesizing all major concepts of Web Systems and Technologies 2 in a socially relevant context.

CHAPTER 2: MINIMUM VIABLE PRODUCT (MVP) DOCUMENTATION

2.1 System Architecture

The SAGANA system follows a three-tier client-server architecture built on the MERN stack, deployed entirely on cloud infrastructure. The React.js frontend, built with Vite, runs in the user's browser and communicates exclusively with the backend through RESTful HTTP requests. The Node.js and Express.js backend serves as the central application layer, processing all business logic, enforcing authentication and role-based access control, and acting as the sole interface between the frontend and the MongoDB Atlas database. MongoDB Atlas is hosted on a cloud cluster, ensuring high availability and automated backups without requiring on-premise server management.

When a user interacts with the React frontend, whether logging in, viewing inventory, placing an order, or managing listings, the frontend sends an HTTP request to the Express.js API. The API validates the request, checks the user's JWT token for authentication, enforces RBAC middleware to confirm role permissions, and then performs the required database operation via Mongoose. The result is returned as a JSON response, which the React frontend renders dynamically without requiring a full page reload.

The system is deployed across two cloud platforms: the React frontend is hosted on Vercel, providing global CDN distribution and instant deployments from GitHub, while the Node/Express backend is hosted on Render, offering a free-tier Node.js server environment with automatic HTTPS. Both platforms are connected to the same MongoDB Atlas cluster via environment variables, ensuring the system operates as a unified, cloud-native application.

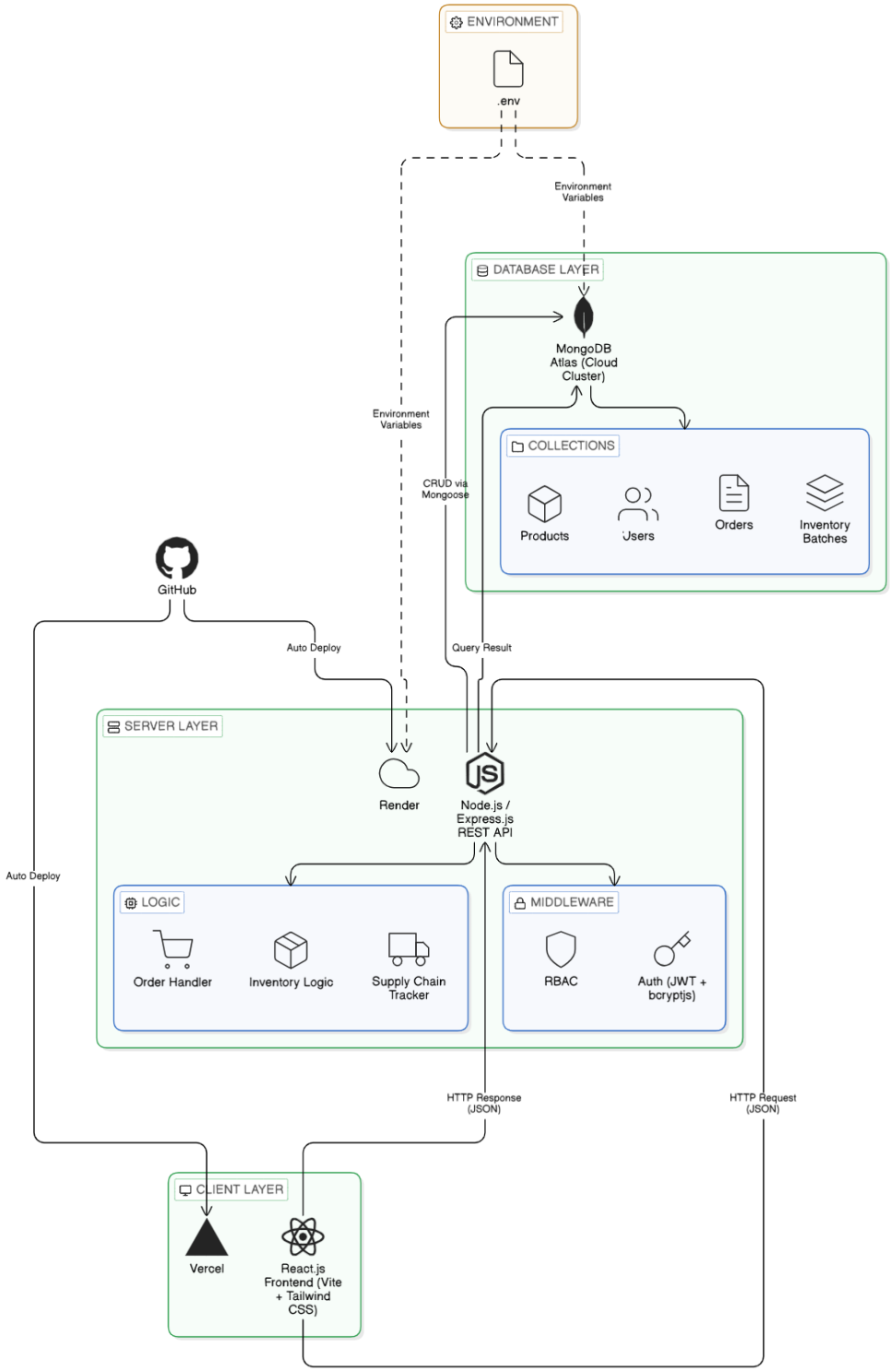

Figure 1: Sagana System Architecture

2.2 Tech Stack & Tools

1. **Frontend**: React.js (v18), Vite, React Router v6, Tailwind CSS, Framer Motion, Recharts
2. **Backend**: Node.js, Express.js, Mongoose ODM
3. **Database**: MongoDB Atlas (Cloud NoSQL), Mongoose for schema validation
4. **Authentication**: JSON Web Tokens (JWT), bcryptjs for password hashing
5. **Testing**: Jest and Supertest for unit and integration tests; Postman and Newman for E2E API testing
6. **Version Control**: Git and GitHub (monorepo with /client and /server folders)
7. **Deployment**: Vercel (frontend), Render (backend), MongoDB Atlas (database)
8. **Environment Management**: .env files for all sensitive credentials; never committed to version control

2.3 Database Schema (Data Modeling)

The SAGANA system uses four primary MongoDB collections. Relationships between collections are implemented using MongoDB References (ObjectId), not embedding, to prevent data duplication and allow efficient querying across collections.

Collection 1: Users

Stores authentication credentials and role assignments for all system users.

Field	Type	Validation / Notes
_id	ObjectId	Auto-generated primary key
name	String	Required, trimmed
email	String	Required, unique, lowercase
password	String	Required, hashed via bcryptjs (min 8 chars)
role	String	Enum: 'admin', 'farmer', 'buyer' — default: 'buyer'
createdAt	Date	Auto-generated via timestamps: true

Collection 2: Products

Contains the metadata for each agricultural product listed by a Farmer Member, pending admin approval before going live in the marketplace.

Field	Type	Validation / Notes
_id	ObjectId	Auto-generated primary key
farmer	ObjectId	References Users._id (Farmer role)
name	String	Required — crop/product name

category	String	Enum: 'Vegetable', 'Fruit', 'Grain', 'Spice', 'Fresh Meat', 'Seafood', 'Other'
description	String	Optional product details
pricePerUnit	Number	Required, min: 0
unit	String	e.g., 'kg', 'bundle', 'piece'
isApproved	Boolean	Default: false — admin must approve before listing
createdAt	Date	Auto-generated via timestamps: true

Collection 3: Inventory Batches

Tracks specific harvest lots from the moment of logging through storage to marketplace listing. Each batch is linked to a Product and the Farmer who harvested it.

Field	Type	Validation / Notes
_id	ObjectId	Auto-generated primary key
product	ObjectId	References Products._id
farmer	ObjectId	References Users._id (Farmer role)
quantity	Number	Required, min: 0 — current stock level
harvestDate	Date	Required
expiryDate	Date	Required — triggers low-stock/expiry alerts
status	String	Enum: 'Harvested', 'Stored', 'Listed', 'Sold', 'Expired'
batchCode	String	Auto-generated unique identifier for traceability

Collection 4: Orders

Manages the full transaction lifecycle between a Buyer and the cooperative marketplace. Each order references both the Product and the Buyer.

Field	Type	Validation / Notes
_id	ObjectId	Auto-generated primary key
buyer	ObjectId	References Users._id (Buyer role)
product	ObjectId	References Products._id
inventoryBatch	ObjectId	References InventoryBatches._id
quantityOrdered	Number	Required — validated against available batch stock

totalPrice	Number	Computed: quantityOrdered x pricePerUnit
status	String	Enum: 'Pending', 'Confirmed', 'Completed', 'Cancelled'
createdAt	Date	Auto-generated via timestamps: true

CHAPTER 3: REST API DESIGN DOCUMENTATION

3.1 Overview & Intent

- **Base URL (Development)**: http://localhost:5000/api/v1
- **Base URL (Production)**: https://sagana.onrender.com/api/v1
- **Purpose**: The SAGANA REST API exposes secure endpoints for managing Users, Products, Inventory Batches, and Orders. It enforces role-based access control on all protected routes and returns standardized JSON responses for both success and error states.
- **Data Format**: All requests and responses use JSON. All protected routes require the Authorization: Bearer <token> header.

3.2 Authentication & Authorization

Strategy: JSON Web Tokens (JWT) are issued upon successful login and must be passed via the Authorization: Bearer <token> header on all protected routes. Tokens expire after 7 days. Passwords are hashed using bcryptjs with a salt round of 10 before storage.

Role	Access Level	Permitted Actions
admin	Full system access	Manage all users, approve/reject products, view all orders and inventory
farmer	Own records only	Log inventory batches, manage own products, view own order status
buyer	Public marketplace	Browse approved products, place and track orders

3.3 Endpoint Documentation

Auth Routes — /api/v1/auth

Method	Endpoint	Auth Required	Description
POST	/auth/register	None	Register a new user (buyer by default)
POST	/auth/login	None	Login and receive a JWT token
GET	/auth/me	Any logged-in user	Get the current user's profile
GET	/auth/users	Admin only	Get all registered users (Admin dashboard)

POST /auth/register — Request Body

```
{
 "name": "Juan Dela Cruz",
```

```
 "email": "juan@torrijos.coop",
 "password": "securepass123",
 "role": "farmer"
}
```

POST /auth/register — Success Response (201 Created)

```
{
 "success": true,
 "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
 "user": { "_id": "64a1...", "name": "Juan Dela Cruz", "role": "farmer" }
}
```

POST /auth/register — Error Response (400 Bad Request)

```
{
 "success": false,
 "message": "Email already in use"
```

Product Routes — /api/v1/products

Method	Endpoint	Auth Required	Description
GET	/products	None (public)	Get all approved products (marketplace)
GET	/products/:id	None (public)	Get a single product by ID
POST	/products	Farmer	Create a new product listing (pending approval)
PUT	/products/:id	Farmer (own) / Admin	Update a product
PUT	/products/:id/approve	Admin only	Approve a product for the marketplace
DELETE	/products/:id	Admin only	Delete a product
GET	/products/all	Admin only	Get ALL products including unapproved (Admin dashboard)

POST /products — Request Body

```
{
 "name": "Pechay",
 "category": "Vegetable",
 "description": "Freshly harvested Pechay from Torrijos farms",
 "pricePerUnit": 35,
 "unit": "kg"
}
```

POST /products — Success Response (201 Created)

```
{
 "success": true,
 "data": {
  "_id": "64b2...", "name": "Pechay", "isApproved": false,
  "farmer": "64a1...", "pricePerUnit": 35, "unit": "kg"
 }
```

Inventory Batch Routes — /api/v1/inventory

Method	Endpoint	Auth Required	Description
GET	/inventory	Admin / Farmer (own)	Get all inventory batches
GET	/inventory/:id	Admin / Farmer (own)	Get a single batch by ID
POST	/inventory	Farmer	Log a new harvest batch
PUT	/inventory/:id	Farmer (own) / Admin	Update batch status or quantity
DELETE	/inventory/:id	Admin only	Delete a batch record
GET	/inventory/available	Public	Get available batches for stock status display (guests and buyers)
GET	/inventory/analytics	Farmer only	Get predictive analytics for a specific crop — returns monthlyYields (real data) and predictiveScores (0-100 smoothed score with seasonal adjustments)

POST /inventory — Request Body

```
{
 "product": "64b2...",
 "quantity": 150,
 "harvestDate": "2026-03-20",
 "expiryDate": "2026-04-05",
 "status": "Harvested"
}
```

POST /inventory — Success Response (201 Created)

```
{
 "success": true,
 "data": {
  "_id": "64c3...", "batchCode": "BCH-20260320-001",
  "quantity": 150, "status": "Harvested"
 }
}
```

Order Routes — /api/v1/orders

Method	Endpoint	Auth Required	Description
GET	/orders	Admin	Get all orders system-wide
GET	/orders/my	Buyer	Get the current buyer's order history
POST	/orders	Buyer	Place a new order

PUT	/orders/:id/status	Admin	Update order status (Confirm / Complete / Cancel)
PUT	/orders/:id	Buyer (own order)	Edit order quantity (Buyer can update their own pending order)
DELETE	/orders/:id	Buyer (own order)	Delete own order (auto-syncs removal from Admin Dashboard)

POST /orders — Request Body

```
{
 "product": "64b2...",
 "inventoryBatch": "64c3...",
 "quantityOrdered": 10
}
```

POST /orders — Success Response (201 Created)

```
{
 "success": true,
 "data": {
  "_id": "64d4...", "status": "Pending",
  "totalPrice": 350, "quantityOrdered": 10
 }
}
```

POST /orders — Error Response (400 Bad Request)

```
{
 "success": false,
 "message": "Insufficient stock in selected batch"
}
```

3.4 Advanced API Features

The GET /products and GET /inventory endpoints support query parameters for pagination and filtering:

```
GET /api/v1/products?category=Vegetable&page=1&limit=10
GET /api/v1/inventory?status=Stored&farmer=64a1...
```

Query Parameter	Applies To	Example	Description
category	Products	?category=Fruit	Filter by product category
status	Inventory, Orders	?status=Pending	Filter by current status
page	Products, Inventory	?page=2	Page number for pagination (default: 1)
limit	Products, Inventory	?limit=10	Items per page (default: 10, max: 50)

sort	Products	?sort=pricePerUnit	Sort results by field (prefix - for descending)

API Versioning Strategy

All endpoints are prefixed with /api/v1/ to support future breaking changes without disrupting existing consumers. If a future version introduces incompatible changes, /api/v2/ will be released while /v1/ remains functional for a defined deprecation period.

3.5 Testing Tools & Examples

- **Postman**: Used for manual endpoint testing during development. A Postman Collection is maintained in the /docs folder of the repository, containing pre-configured requests for all endpoints with environment variables for base URL and token.
- **Newman**: Used for automated CLI-based E2E testing. The Postman Collection is exported and run via Newman in sequence to validate the full user flow: Register → Login → Create Product → Approve Product → Log Inventory → Check Stock (/inventory/available) → Place Order → Edit Order → Delete Order → Update Order Status → View Analytics (/inventory/analytics).
- **Jest + Supertest**: Used for unit and integration testing of the Express.js backend. Each route handler has corresponding test files that verify correct status codes, response shapes, and error handling behavior.

Sample Frontend Fetch Request

```
// Fetch approved products from the public marketplace
const res = await fetch('https://sagana.onrender.com/api/v1/products', {
  method: 'GET',
  headers: { 'Content-Type': 'application/json' }
});
const data = await res.json();
```

Sample Authenticated Request

```
// Place an order (requires JWT token)
const res = await fetch('https://sagana.onrender.com/api/v1/orders', {
  method: 'POST',
  headers: {
    'Content-Type': 'application/json',
    'Authorization': `Bearer ${token}`
  },
  body: JSON.stringify({ product: productId, inventoryBatch: batchId, quantityOrdered: 5 })
});
```

CHAPTER 4: DEPLOYMENT & LINKS

4.1 Live Application Links

Resource	URL / Link

Frontend Web Application	https://sagana-mu.vercel.app
Backend API Base URL	https://sagana.onrender.com
GitHub Master Repository	https://github.com/Drhylicious/SAGANA

4.2 Testing Coverage Summary

Unit & Integration Tests

The following table summarizes the expected Jest code coverage targets for the SAGANA backend. Actual coverage results will be pasted here upon completion of the test suite.

```
-------------------------|---------|----------|---------|---------|-------------------------------------------------------------
File                     | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------------|---------|----------|---------|---------|-------------------------------------------------------------
All files                |   58.57 |    28.81 |      50 |   59.57 |
 controllers             |   46.53 |    23.63 |      48 |   47.45 |
  authController.js      |   87.87 |       75 |     100 |   87.87 | 41,55,70,82
  inventoryController.js |    37.5 |    13.63 |   33.33 |   37.73 | 23,31-47,69,77-95,103-111,119-134
  orderController.js     |   68.62 |       50 |      75 |   69.38 | 21,35,41,66,79,89,94,102-119
  productController.js   |   27.61 |    10.34 |   27.27 |   28.71 | ...,25-61,83,91-98,110,126,136-170,185,189,197-212,220-233
 middleware              |   94.44 |      100 |     100 |   94.44 |
  authMiddleware.js      |   91.66 |      100 |     100 |   91.66 | 13
  roleMiddleware.js      |     100 |      100 |     100 |     100 |
 models                  |     100 |      100 |     100 |     100 |
  InventoryBatch.js      |     100 |      100 |     100 |     100 |
  Order.js               |     100 |      100 |     100 |     100 |
  Product.js             |     100 |      100 |     100 |     100 |
  User.js                |     100 |      100 |     100 |     100 |
 routes                  |   86.44 |      100 |       0 |   86.44 |
  authRoutes.js          |   66.66 |      100 |       0 |   66.66 | 13-18
  inventoryRoutes.js     |     100 |      100 |     100 |     100 |
  orderRoutes.js         |     100 |      100 |     100 |     100 |
  productRoutes.js       |   86.36 |      100 |       0 |   86.36 | 7-11
-------------------------|---------|----------|---------|---------|-------------------------------------------------------------
Test Suites: 1 passed, 1 total
Tests:       25 passed, 25 total
Snapshots:   0 total
Time:        13.72 s, estimated 15 s
Ran all test suites.
```

End-to-End (E2E) Tests

The automated Postman workflow tests the full cooperative user journey in sequence, validating that all API endpoints function correctly as an integrated system. The Newman CLI runner is used to execute the collection automatically.

Step	Action	Expected Result
1	POST /auth/register — Register Admin	201 Created, JWT token returned
2	POST /auth/register — Register Farmer	201 Created, JWT token returned
3	POST /auth/register — Register Buyer	201 Created, JWT token returned
4	POST /auth/login — Farmer Login	200 OK, valid JWT token

5	POST /products — Farmer creates Product	201 Created, isApproved: false
6	PUT /products/:id/approve — Admin approves	200 OK, isApproved: true
7	POST /inventory — Farmer logs harvest batch	201 Created, batchCode assigned
8	GET /products — Buyer views marketplace	200 OK, approved products returned
9	POST /orders — Buyer places order	201 Created, status: Pending
10	PUT /orders/:id/status — Admin confirms	200 OK, status: Confirmed
11	GET /inventory/available — Guest checks stock status	200 OK, available batches with stock returned
12	GET /inventory/analytics — Farmer views predictive analytics	200 OK, monthlyYields and predictiveScores (0-100) returned
13	PUT /orders/:id — Buyer edits order quantity	200 OK, updated quantityOrdered and totalPrice returned
14	DELETE /orders/:id — Buyer deletes own order	200 OK, order removed and auto-synced from Admin Dashboard

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APPENDICES

Appendix A: Glossary of API Terms

Term	Definition
API (Application Programming Interface)	A set of rules and protocols that allows different software applications to communicate with each other. SAGANA uses a REST API to connect the React frontend and the Node.js backend.
Endpoint	A specific URL in an API that accepts requests and returns responses. For example, POST /api/v1/auth/login is an endpoint for user authentication.
HTTP Method	The type of operation an API request performs. The four used in SAGANA are GET (retrieve), POST (create), PUT (update), and DELETE (remove).
JSON (JavaScript Object Notation)	A lightweight, human-readable format for transmitting data between a client and a server. All SAGANA API requests and responses use JSON format.
JWT (JSON Web Token)	A compact, digitally-signed token used to securely transmit authentication information. SAGANA issues a JWT upon login, which the client includes in subsequent requests to prove identity.
Bearer Token	An authorization scheme where the client includes a JWT in the HTTP header as: Authorization: Bearer <token>. Required for all protected SAGANA endpoints.
Payload	The actual data sent in the body of an API request. For example, the payload of a POST /products request includes the product name, category, and price.
CRUD	An acronym for Create, Read, Update, and Delete — the four basic operations performed on database records. SAGANA implements full CRUD for all four collections.
RBAC (Role-Based Access Control)	A security model that restricts system access based on the user's assigned role. SAGANA uses three roles: admin, farmer, and buyer, each with distinct permissions.
Middleware	A function in Express.js that runs between receiving a request and sending a response. SAGANA uses middleware for JWT verification and role checking on protected routes.
Status Code	A standardized numerical code in an HTTP response indicating the outcome of a request (e.g., 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Internal Server Error).
Schema Validation	The process of enforcing data structure rules on incoming data before it is stored in the database. SAGANA uses Mongoose schema validation to prevent corrupt or incomplete records.

Appendix B: Acronyms & Abbreviations

Acronym	Full Form
API	Application Programming Interface
BFAR	Bureau of Fisheries and Aquatic Resources
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DA	Department of Agriculture
DTI	Department of Trade and Industry
E2E	End-to-End (Testing)
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
JWT	JSON Web Token
JSON	JavaScript Object Notation
LGU	Local Government Unit
SP3	Samahan ng Pagkakaisa sa Pag-unlad ng Payanas (SP3) Agriculture Cooperative
MERN	MongoDB, Express.js, React.js, Node.js
MVP	Minimum Viable Product
NoSQL	Non-relational (Not Only SQL) Database
ODM	Object Document Mapper
RBAC	Role-Based Access Control
REST	Representational State Transfer
SAGANA	Supply chain, Agricultural Governance, and Agricultural Network Application
SDG	Sustainable Development Goal
UPLB	University of the Philippines Los Banos
USAID	United States Agency for International Development
URL	Uniform Resource Locator
Vite	A modern frontend build tool for React applications

Appendix C: Changelog / Document Revisions

The following table records all major revisions made to this project proposal document throughout the semester.

Version	Date	Author(s)	Changes Made
v1.0	March 19, 2026	Salangsang, J.D.	Initial proposal draft submitted — Chapter 1 complete
v1.1	March 25, 2026	All Members	Updated project scope from Web and Mobile-Based to Web-Based; revised title throughout
v1.2	March 28, 2026	All Members	Added Chapters 2, 3, 4 and Appendices A–D
v1.3	March 28, 2026	All Members	Updated Section 1.5 with SP3 certification; updated location from Boac to Torrijos; added new API endpoints (/inventory/analytics, /inventory/available, /orders buyer edit & delete)
v2.0	April 13, 2026	All Members	Final version submitted — all deployment links, predictive analytics (Option 2 — Real MongoDB Data), buyer order management (edit & delete), and v2.1 features fully documented

Appendix D: Proof of Validity / Certification

This appendix contains the signed Certification of Project Validity and Endorsement issued by the SP3 Agriculture Cooperative of Torrijos, Marinduque, under the office of SP3 Agriculture Cooperative leadership, and noted by Jenny Reanzares, Samahan ng Pagkakaisa sa Pag-unlad ng Payanas (SP3) Agriculture Cooperative.

The certification confirms that the problems identified in the SAGANA study accurately reflect the operational challenges experienced by the Payanas Cooperative of Torrijos, Marinduque, and that the proposed system is endorsed as academically and practically relevant.

Republic of the Philippines
Province of Marinduque

MUNICIPALITY OF TORRIJOS

Samahan ng Pagkakaisa sa Pag-unlad ng Payanas (SP3) Agriculture Cooperative
Torrijos, Marinduque, Philippines

The Research Panel
BSI/T-SD 3D, College of Information and Computing Sciences
Marinduque State University
Torrijos, Marinduque

Subject: Certification of Project Validity and Endorsement

To Whom It May Concern,

This is to certify that the Samahan ng Pagkakaisa sa Pag-unlad ng Payanas (SP3) Agriculture Cooperative of Torrijos, Marinduque has been consulted by the following Bachelor of Science in Information Technology (BSI/T) students of Marinduque State University regarding their final project proposal entitled:

SAGANA: A Web-Based Integrated Agricultural Supply Chain, Inventory, and Digital Marketplace System for the Cooperative of Payanas, Torrijos, Marinduque

Proponents:

1. Jhon Drhy M. Salangsang — Leader & Full Stack Developer
2. John Harvey Revilloza — Backend Developer
3. John Harold Revilloza — QA / Tester
4. Marc Louie M. Yañez — Frontend Developer

After reviewing the project description and objectives presented by the proponents, this office hereby certifies the following:

1. The problems identified in the study, including the reliance on manual and paper-based systems for inventory tracking, supply chain management, and market transactions in the agricultural cooperatives of Torrijos, Marinduque, are accurate and reflect the actual operational challenges experienced by local cooperative organizations.
2. The proposed system, SAGANA, addresses a genuine and pressing need within the agricultural sector of the municipality. The integration of a digital inventory system, supply chain tracker, and direct-to-consumer marketplace is aligned with the digitalization goals of the Department of Agriculture and the local government unit.
3. This office endorses the project as academically and practically relevant, and recognizes its potential to improve the operational efficiency, market access, and income generation capacity of the cooperatives of Torrijos, Marinduque.
4. The proponents are authorized to reference this office and the information gathered during their consultations for the purpose of their academic study. All information shared has been provided voluntarily in support of their research.

This document serves as Appendix D — Proof of Validity / Certification — as required in Section 1.5 of the SAGANA Final Project Proposal submitted to the College of Information and Computing Sciences, Marinduque State University.

Republic of the Philippines
Province of Marinduque

MUNICIPALITY OF TORRIJOS

Samahan ng Pagkakaisa sa Pag-unlad ng Payanas (SP3) Agriculture Cooperative
Torrijos, Marinduque, Philippines

This certification is issued upon the request of the proponents and for whatever legal and academic purposes it may serve.

Respectfully yours,

JOHN HAROLD REVILLOZA

JOHN HARVEY REVILLOZA

JHON DRIFY M. SALANGSANG

MARC LOUIE M. YAÑEZ

Certified and validated by:

JENNY REANZARES
BOD Chairperson
Samahan ng Pagkakaisa sa Pag-unlad ng Payanas (SP3) Agriculture Cooperative

This document serves as Appendix D – Proof of Validity / Certification — as required in Section 1.5 of the SAGANA Final Project Proposal submitted to the College of Information and Computing Sciences, Marinduque State University